

DRAFT — DBA-ES-GN-FC8: Renewable Generation and BESS

Renewable Generation and Battery Energy Storage

Utility-Scale Solar, Wind, and BESS

Design Basis Accident — Military Action

DBA-ES-GN-FC8 — Facility Class Document

Issued Under: DBA-ES Electric Sector Framework

Version 0.1 — DRAFT

June 2026

Timothy E. Roxey

Eclectic Technologies

Prepared for: Electric Sector — Industry, Government, and Allied Partners

Parent document: DBA-ES Electric Sector Framework v0.7

Preface

Every generation technology addressed in the DBA-ES-GN series so far — coal, oil-fired — relies on a supply chain whose geographic concentration creates strategic risk but whose physical components are well-understood, domestically manufactured in significant part, and whose failure modes are governed by decades of operational experience. Utility-scale solar, wind, and battery energy storage are different in every one of these respects. Their supply chains are concentrated in a single foreign country at a scale that has no precedent in the U.S. energy infrastructure history. Their critical components — inverters, battery management systems, solar panels, and rare-earth permanent magnets — are manufactured by firms operating under the jurisdiction of a government that has demonstrated the intent and capability to embed undisclosed functionality in infrastructure hardware. And their failure modes under adversarial manipulation are not governed by decades of operational experience; they are governed by the theoretical capabilities of power electronics firmware running on digital signal processors whose source code is not available for independent security review.

This document establishes the Design Basis Accident scenarios for deliberate military action against utility-scale renewable generation and battery energy storage infrastructure. The facility class encompasses utility-scale solar photovoltaic clusters, onshore and offshore wind generation clusters, and utility-scale battery energy storage systems (BESS). The three technologies are addressed in a single instrument because they share a governing analytical characteristic that distinguishes FC8 from every other DBA-ES-GN instrument: the primary attack surface is not the physical plant or the SCADA system. It is the generation equipment itself — specifically the inverter, whose firmware governs the electrical behavior of every inverter-based resource (IBR) on the grid and which, if adversarially compromised, can produce grid-destabilizing behavior from thousands of devices simultaneously without any kinetic action, without any SCADA intrusion, and without leaving any forensic evidence at the facility.

The GenMix Optimization Whitepaper (v0.3) established the technical foundation for this analysis: Chinese manufacturers control approximately 80 to 85 percent of global utility inverter production, approximately 80 percent of global solar panel manufacturing, and approximately 85 to 90 percent of rare earth permanent magnet processing. CISA and the Department of Energy have documented undisclosed remote access functionality in Chinese-manufactured inverters deployed in U.S. grid infrastructure. The inverter firmware attack is not a theoretical scenario. The attack surface is already embedded in the generation fleet, at a scale whose consequence potential has not been formally assessed in any existing regulatory framework.

The evidentiary anchors for the physical consequence chains in FC8 are the February 2021 Winter Storm Uri solar cluster zero-output event in ERCOT — which demonstrated how geographically correlated weather events can simultaneously remove large fractions of solar generation capacity — the pattern of multi-day wind droughts documented in NERC's long-term

reliability assessments, and the 2023 Moss Landing BESS fire — the largest BESS fire in U.S. history at that time — which demonstrated that lithium battery thermal runaway in a large-scale storage facility produces consequences that overwhelm local fire suppression resources and may require facility abandonment.

1. Introduction

1.1 Purpose

This document establishes the Design Basis Accident — Military Action scenarios for the renewable generation and battery energy storage facility class. Its purpose is to define the bounding threat event that utility-scale solar, wind, and BESS security planning, resilience investments, and recovery architecture must address. As with all DBA-MA documents, this document does not prescribe specific protective measures at individual facilities. It defines the consequence envelope within which protective measures must perform.

A renewable generation security program designed against conventional physical threats will fail when a sophisticated adversary exploits the supply chain attack surface that is already embedded in the fleet — compromised inverter firmware that produces synchronized grid-destabilizing behavior from thousands of devices on command. Physical hardening of solar panels, wind turbines, and BESS enclosures does not address the firmware attack vector. The DBA-MA approach must account for both the physical destruction scenarios and the firmware attack as coequal Design Basis events.

1.2 Scope and Applicability

This document addresses utility-scale renewable generation and battery energy storage facilities at the cluster level. Applicability criteria: utility-scale solar PV clusters at or above 200 MW aggregate AC nameplate capacity; onshore wind clusters at or above 200 MW aggregate nameplate capacity; offshore wind arrays at or above 200 MW aggregate nameplate capacity; and utility-scale BESS installations at or above 100 MW / 400 MWh. The cluster-level threshold reflects the consequence-chain organizing principle: individual renewable generation units are individually small-consequence; it is the geographic clustering of IBR capacity that produces the Tier 2 grid-level consequence when a large block of capacity simultaneously becomes unavailable.

Distributed solar and distributed BESS at the distribution level are addressed in DBA-ES-DI-FC5 as edge generation components. This document addresses bulk power system-connected utility-scale resources only. The inverter supply chain risk addressed in Section 3.2 applies to both utility-scale and distributed IBR installations. Still, the grid consequence analysis in this document is scoped to the bulk power system impact.

1.3 Relationship to Governing Documents

This document is issued under the DBA-ES Electric Sector Framework (v0.7), which is itself issued under the DBA-EN-SF Energy Sector Framework (v0.3) and the DBA-MA Parent Framework (IAEA L1 v1.4). The GN type code and FC numbering convention are established in DBA-ES-GN-FC6. The GenMix Optimization Whitepaper (v0.3) is a foundational reference document for the RT_HVLLC values, the inertia constant ($H = 0$) characterization of IBRs, and the supply chain concentration data that governs the Scenario A analysis. The Bright Line and Command Broker requirements are inherited unchanged and applied in Section 3.3 with particular attention to the inverter firmware supply chain dimension.

Offshore wind arrays share the maritime access and recovery constraint framework established in DBA-OIL-FC2 for offshore crude production. This document references that framework for offshore wind recovery without fully reproducing the maritime operations analysis; a dedicated offshore wind use case instrument may follow at a future revision.

1.4 The IBR Triad — Three Technologies, One Governing Vulnerability

Utility-scale solar, wind, and BESS are three distinct technologies with distinct generation profiles, distinct physical vulnerability profiles, and distinct recovery timelines. They share a single governing vulnerability: the inverter. All three technologies produce AC electricity for the grid through power electronics inverters that synthesize output current at commanded power and frequency. The inverter's firmware governs the electrical behavior of the device — its response to grid frequency deviations, its contribution to or withdrawal from voltage support, its protection settings, and, in grid-forming devices, its synthetic inertia behavior.

An adversary who has embedded malicious firmware in inverters across all three technology types — solar, wind, and BESS — has embedded an attack surface across the full IBR fleet simultaneously. A synchronized command to all compromised inverters to withdraw from the grid, to inject destabilizing reactive current, or to cease synthetic inertia response at the moment of a grid frequency event produces a compound consequence whose scale is proportional to the fleet's total installed capacity, not to any individual facility's size. This is the attack that has no precedent in the DBA-ES series and no analog in any prior generation technology: a single adversarial action that produces simultaneous consequences across thousands of geographically dispersed facilities.

2. Facility Profile and Asset Taxonomy

2.1 Utility-Scale Solar PV

Utility-scale solar PV clusters consist of thousands to millions of individual solar panels arranged in tracking or fixed-tilt arrays, connected through string inverters or central inverters to the collection system and ultimately to the grid interconnection transformer. Modern utility-scale

solar plants use string inverters — individual inverters serving strings of 20 to 40 panels — rather than central inverters, improving granular fault detection but increasing the total inverter count per facility by orders of magnitude. A 200 MW solar facility may contain 2,000 to 5,000 individual string inverters.

Solar PV generation is deterministically zero for approximately half of every 24 hours and substantially reduced during low-sun-angle periods. Its geographic correlation with weather means that large solar clusters within the same regional climate zone lose output simultaneously during cloud cover events, dust storms, and icing events. The February 2021 Uri event in ERCOT demonstrated this: iced solar panels across the Texas Gulf Coast contributed to approximately 16 GW of forced outage during the critical demand period, representing a geographically correlated simultaneous generation loss with no single-facility analog.

2.2 Utility-Scale Wind — Onshore

Onshore wind clusters consist of individual wind turbine generators (WTGs) — typically 3 to 6 MW each — arranged in rows or clusters optimized for wind resource and wake avoidance. Each WTG contains a nacelle assembly housing the generator, gearbox (in some designs), power electronics, and control systems. The nacelle is the highest-consequence single component from an RT_HVLLC perspective: gearbox replacement or major nacelle rebuild requires 6 to 12 months, consistent with the Table 1 value from the GenMix Optimization Whitepaper. Modern direct-drive turbines eliminate the gearbox but introduce rare-earth permanent magnet generators whose supply chain is approximately 85 to 90 percent Chinese-controlled at the processing stage.

Wind generation is subject to multi-day calm periods — wind droughts — that can persist for three or more consecutive days across wide geographic areas. Unlike the solar cluster zero-output event, which is weather-correlated and predictable, wind droughts are harder to forecast beyond 3 to 5 days. They can affect multiple regional wind resource areas simultaneously during large anticyclonic weather patterns. The governing non-adversarial wind bounding event is a 5-day calm period affecting a major wind resource zone during a high-demand period when wind has been relied upon in the generation dispatch plan.

2.3 Utility-Scale Wind — Offshore

Offshore wind arrays — currently concentrated in the Northeast and Mid-Atlantic coastal waters, with development expanding across the Gulf of Mexico and West Coast — face the same inverter supply chain vulnerability as onshore wind but add the maritime access dimension documented in DBA-OIL-FC2. Offshore wind turbine maintenance and repair require specialized offshore construction vessels and crew transfer vessels whose availability is limited and whose mobilization requires days to weeks. During high sea states, offshore access is physically impossible regardless of available vessel capacity. An adversary who targets offshore wind nacelles or foundation structures selects a target whose physical recovery combines the onshore wind RT_HVLLC (6 to 12 months for major nacelle components) with the offshore

construction vessel constraint (weeks to months for vessel mobilization) established in DBA-OIL-FC2.

Offshore wind arrays also share the subsea cable vulnerability identified in DBA-OIL-FC2 for export pipelines: the inter-array cables and the export cable connecting the offshore array to the onshore grid are subsea assets whose physical repair requires specialized cable-lay vessels. A coordinated attack on both the turbine nacelles and the export cable produces a compound recovery requirement that exceeds offshore wind's normal maintenance infrastructure.

2.4 Utility-Scale BESS

Utility-scale battery energy storage systems consist of battery modules — typically lithium iron phosphate (LFP) or nickel manganese cobalt (NMC) chemistry — housed in containerized enclosures connected through battery management systems (BMS) and power conversion systems (PCS, which are inverters) to the grid. A 100 MW / 400 MWh BESS installation may consist of hundreds of individual battery containers, each managed by a BMS that monitors cell temperature, state of charge, and cell voltage balance.

BESS is the only IBR technology in FC8 that provides dispatchable generation — it can discharge on command regardless of weather conditions, making it the renewable sector's equivalent of the oil-fired peaker in DBA-ES-GN-FC7. As BESS installations grow in scale and grid importance, they increasingly serve the same RMR functions as oil-fired peakers — providing fast-response frequency regulation, voltage support, and emergency capacity during grid stress events. Their vulnerability profile combines the inverter/firmware attack surface (PCS inverters are subject to the same supply chain concentration as solar and wind inverters) with the thermal runaway risk unique to lithium chemistry battery systems.

The 2023 Moss Landing BESS fire — at a facility rated at approximately 182.5 MW / 730 MWh — involved thermal runaway propagation through multiple battery containers, producing a fire that burned for days and required firefighters to allow the fire to burn down rather than risk firefighter casualties in attempting suppression. The facility sustained total loss. The governing BESS non-adversarial bounding event is the Moss Landing fire profile applied to a facility scaled to the FC8 applicability threshold.

2.5 The Inverter as the Common Attack Surface

The power electronics inverter is the single component that is present in all three FC8 technology types — solar string inverters and central inverters, wind turbine full-converter and doubly-fed induction generator controllers, and BESS power conversion systems — and whose firmware governs the electrical behavior of each device on the grid. The supply chain concentration established in the GenMix Optimization Whitepaper is the governing context: Chinese manufacturers control approximately 80 to 85 percent of global utility inverter production, and

CISA has documented undisclosed remote access functionality in Chinese-manufactured inverters deployed in U.S. grid infrastructure.

The inverter firmware attack surface has three distinct attack pathway types. First, pre-installation firmware compromise: inverters shipped from the manufacturer with malicious firmware already installed, designed to activate on a remote command or on a specific trigger condition. Second, over-the-air firmware update compromise: legitimate firmware update channels are exploited to push malicious firmware to already-deployed inverters. Third, direct communication channel exploitation: undisclosed communication interfaces in deployed inverters are used to issue malicious commands without any firmware modification. All three pathways produce the same consequence: coordinated adversarial control of thousands of grid-connected power electronics devices.

2.6 RT_HVLLC Values — FC8 Equipment

Technology / Component	RT_HVLLC	Governing Constraint	Source
Solar PV — string inverter replacement	3–6 months	Manufacturing queue; Chinese supply chain concentration	GenMix Whitepaper v0.3 Table 1
Solar PV — panel replacement (large cluster)	4–8 months	Logistics: ~80% Chinese manufacture	GenMix Whitepaper v0.3
Onshore wind — nacelle/gearbox major repair	6–12 months	Crane mobilization; component fabrication	GenMix Whitepaper v0.3 Table 1
Offshore wind — nacelle/gearbox major repair	12–24 months	Offshore vessel availability; weather windows; maritime access (DBA-OIL-FC2)	GenMix Whitepaper v0.3 + DBA-OIL-FC2
BESS — full battery container replacement (thermal runaway loss)	6–12 months	Battery module manufacturing queue; site remediation before re-installation	GenMix Whitepaper v0.3 Table 1
BESS — PCS/inverter replacement	3–6 months	Same supply chain as solar string inverters	GenMix Whitepaper v0.3

2.7 Regulatory Authority

NERC CIP. Utility-scale IBR facilities meeting BES definitions are subject to NERC CIP cybersecurity standards. CIP-013 supply chain risk management requirements — requiring registered entities to assess cybersecurity risks in their industrial control system supply chains — are the primary regulatory mechanism addressing the inverter firmware attack surface. CIP-013 requirements have not yet been fully calibrated to the specific risk profile of Chinese-manufactured inverters with documented undisclosed functionality.

FERC. FERC Order 841 established the right of BESS to participate in wholesale electricity markets. FERC interconnection standards govern the technical requirements for IBR grid connection, including protection settings and low-voltage ride-through requirements. FERC's authority over inverter firmware security has not been explicitly exercised.

BOEM. The Bureau of Ocean Energy Management has jurisdiction over offshore wind development on the Outer Continental Shelf, including facility safety and environmental requirements. BOEM's security framework for offshore wind does not currently address the inverter firmware attack vector or the DBA-MA kinetic attack scenarios.

DOE / CISA. DOE's Cybersecurity, Energy Security, and Emergency Response (CESER) office and CISA have both issued warnings and guidance on Chinese-manufactured inverter risks. The regulatory framework for mandatory action in response to these warnings remains incomplete. DOE's Prohibition Order authority under the Executive Order on Securing the U.S. Bulk Power System (EO 13920) provides the legal mechanism for mandatory supply chain action; its application to inverters has been partial.

3. Design Basis: Renewable Generation and BESS Facility Class

3.1 Non-Adversarial Bounding Events

The non-adversarial Design Basis for the renewable generation and BESS facility class inherits the infrastructure failure, natural hazard, and operational event categories from the DBA-ES Sector Framework. This document develops the facility-class-specific consequence chains for three governing non-adversarial bounding events.

Solar cluster zero-output — geographically correlated weather event. The governing non-adversarial bounding event for solar PV is a geographically correlated weather event that simultaneously removes output from a major solar cluster during a high-demand period. The February 2021 Winter Storm Uri event in ERCOT is the reference: iced solar panels across a wide geographic area contributed approximately 16 GW of forced outage. The bounding case involves a major solar cluster region experiencing an extended cloud cover, dust storm, or icing event during a seasonal peak demand period when solar generation had been planned as a significant supply component. The consequence is not a loss of generation capacity in the RT_HVLLC sense — the panels are not destroyed — but a loss of output availability for the duration of the weather event, potentially 2 to 5 days, during which the grid must be managed without the planned solar contribution.

Multi-day wind drought. The governing non-adversarial bounding event for wind is a sustained calm period — wind speeds below cut-in across a major wind resource zone — lasting 3 to 5 or more consecutive days. NERC's long-term reliability assessments document the statistical frequency of multi-day wind droughts across the major U.S. wind resource areas; the bounding event involves a 5-day drought coinciding with a demand peak during which wind generation

had been included in the supply plan. Unlike the solar zero-output event, the wind drought has no immediate physical recovery trigger — the wind must return.

BESS thermal runaway — Moss Landing profile. The governing non-adversarial bounding event for BESS is a thermal runaway cascade — initiated by a cell-level failure that propagates through adjacent battery modules and containers — at a major BESS installation. The 2023 Moss Landing event (182.5 MW / 730 MWh) is the reference: thermal runaway in one container propagated to adjacent containers, producing a multi-day fire that resulted in total facility loss. The bounding case involves a major FC8-scale BESS facility (100 MW / 400 MWh or above) experiencing a full thermal runaway cascade, with the governing consequence being not only the immediate generation loss but the 6 to 12-month site remediation and battery module replacement timeline before the facility can be rebuilt.

3.2 Design Basis Accident — Military Action: Bounding Event

The DBA-MA bounding event for the renewable generation and BESS facility class is a coordinated inverter firmware attack — pre-positioned through supply chain compromise, over-the-air firmware update exploitation, or direct communication channel exploitation — that commands simultaneous grid-destabilizing behavior from compromised IBR devices across a major regional grid footprint, executed during a period of maximum IBR penetration and minimum synchronous generation dispatch when the grid's frequency response capability is at its nadir.

The bounding parameters, inherited from the DBA-ES Sector Framework threat taxonomy:

Adversary tier: Tier 1 state actor with deep supply chain access and the technical capability to develop, distribute, and activate malicious inverter firmware across devices from multiple manufacturers. The Chinese government's demonstrated control over major inverter manufacturers — through ownership structures, regulatory authority, and the National Security Law — provides the mechanism by which a state actor can achieve this supply chain access without requiring direct cyber intrusion into individual facilities.

Modality: Supply chain primary with cyber activation. The malicious firmware is distributed through normal product supply chains — either pre-installed at manufacture or pushed through legitimate firmware update channels — over a period of months to years before activation. The activation command is delivered through the undisclosed communication channels documented by CISA, requiring no intrusion into the facility's operational SCADA system.

Target selection: The IBR fleet as a whole, not any individual facility. The adversary who has embedded firmware in 80 percent of deployed utility inverters does not need to select targets in the conventional sense. They activate the firmware simultaneously across the entire compromised population. Target selection is expressed in the timing and behavioral mode of the activation command.

Timing: A period of maximum IBR penetration — a high-solar, high-wind day when gas-fired generation dispatch is minimal and synchronous inertia is at its lowest — when the grid has the least frequency response capability to absorb a sudden large IBR disconnection. NERC's inertia monitoring data identifies these periods with precision; an adversary with access to published grid operations data can identify the optimal activation window.

Behavioral mode options: The firmware attack offers the adversary multiple consequence options. Simultaneous disconnection of all compromised IBRs — removing a large fraction of total generation from the grid in a single instant — produces an instantaneous frequency excursion. Injection of destabilizing reactive current produces a voltage collapse. Withdrawal of synthetic inertia response during a frequency event — causing grid-forming inverters to behave as grid-following devices at the moment of frequency stress — reduces effective system inertia below the threshold required for successful frequency recovery. The adversary selects the behavioral mode that best serves their strategic objective.

3.3 Dual Design Basis Architecture — The Supply Chain Dimension

Consistent with the DBA-ES Sector Framework and DBA-EN-SF (§4.2), a renewable generation and BESS facility may simultaneously carry a sector-specific physical Design Basis established by this document and an AI Governance Design Basis established under the AI Governance series. The two design bases interact at the DBA level. The Bright Line and Command Broker architecture established in CB-Framework_Cross_Sector v0.1 governs the boundary between the two design bases.

FC8 introduces a supply chain dimension to the Bright Line that has no analog in any prior DBA-ES instrument. For all previous facility classes, the Bright Line governs the boundary between ML systems operating autonomously below it and Gen AI advisory tools requiring Command Broker authorization above it. For FC8, the Bright Line must also govern the firmware trustworthiness boundary: an inverter whose firmware has not been independently verified for the absence of undisclosed functionality is operating in an unknown position relative to the Bright Line — its autonomous behavior may not be deterministic or formally testable in the way that below-Bright-Line ML systems are required to be.

Four FC8-specific AI governance interaction points are identified:

Inverter firmware trustworthiness as a Bright Line precondition. The Bright Line architecture assumes that ML systems operating autonomously below it are deterministic and formally testable. This assumption requires that the firmware governing the inverter's behavior is known, complete, and free of undisclosed functionality. An inverter operating on firmware that has not been independently verified against this standard cannot be treated as a below-Bright-Line autonomous ML system — its behavior under adversarial conditions is unknown. The AI Governance Design Basis must establish that firmware verification against an independent source-code-equivalent standard is a precondition for autonomous inverter operation in any grid-

critical application. Until that verification is completed, the inverter must be treated as operating above the Bright Line — requiring human Command Broker oversight for any operational parameter change — regardless of the deterministic appearance of its normal behavior.

BESS BMS and state-of-charge management ML. Battery management systems that autonomously manage cell-level temperature, state of charge, and charge/discharge rate are ML systems that may operate below the Bright Line when performing deterministic cell-level safety functions within their validated operating envelopes. The adversarial concern for BESS BMS is manipulation of state-of-charge reporting — causing the BMS to report higher available capacity than actually exists, masking a degraded state that makes thermal runaway more likely, or causing the BMS to drive cells into overcharge or deep-discharge conditions that accelerate degradation. The Design Basis requirement is that BMS state-of-charge data is validated against independent physical measurements at defined intervals, and that BMS parameter changes require management of change review equivalent to the OSHA PSM standard applied in DBA-GAS-FC2 for safety-critical control system parameters.

Renewable generation portfolio optimization Gen AI. Gen AI advisory tools for renewable generation portfolio optimization — tools that recommend dispatch sequencing, curtailment decisions, and BESS charge/discharge scheduling across a portfolio of solar, wind, and BESS assets — are above the Bright Line and require Command Broker authorization. In the context of a grid under stress, these tools may recommend BESS discharge rates or wind curtailment sequences whose compound effect on grid frequency or voltage is not visible from the individual facility perspective. The Command Broker must evaluate Gen AI dispatch recommendations in the context of the current grid operating conditions — inertia level, frequency response reserve, and voltage support availability — not merely at the individual facility level.

Grid-forming inverter synthetic inertia control. Grid-forming inverters that provide synthetic inertia — emulating the kinetic energy exchange of a synchronous machine through fast power electronics control — are ML systems operating below the Bright Line when their synthetic inertia algorithm operates within its designed parameters. However, the firmware trustworthiness precondition identified above applies with particular force here: a grid-forming inverter whose firmware has been compromised may withdraw synthetic inertia response at the moment of frequency stress — converting from a stabilizing grid-forming device to a destabilizing grid-following device — in a way that is indistinguishable from normal behavior until the grid frequency event reveals the consequence. The Design Basis requirement is that synthetic inertia capability from inverters whose firmware has not been independently verified is not credited in grid stability analysis and is not relied upon in emergency operating procedures.

4. Design Basis Accident Scenarios

4.1 Scenario A: Inverter Firmware Attack — Simultaneous Grid-Destabilizing Behavior

4.1.1 Initiating Event

Activation of pre-positioned malicious firmware across Chinese-manufactured inverters deployed in utility-scale solar, wind, and BESS facilities in a major regional grid footprint — specifically ERCOT, the Eastern Interconnection Mid-Atlantic region, or the Western Interconnection California/Nevada corridor — during a period of maximum IBR penetration. The activation is executed through the undisclosed communication channel documented by CISA, delivering a simultaneous command to all compromised devices to disconnect from the grid on a coordinated timer.

The activation occurs at 2:00 PM on a clear, moderate-wind spring day when solar generation is at approximately 80 percent of nameplate and wind is providing significant output — IBR penetration in the affected region is at approximately 65 to 75 percent of total generation. Synchronous generation dispatch is minimal; system inertia is at its seasonal nadir. The compromised inverter population represents approximately 60 to 70 percent of the total IBR fleet in the affected region, consistent with the Chinese manufacturer market share.

4.1.2 Progression Sequence

T+0: Simultaneous disconnection of approximately 60 to 70 percent of IBR generation in the affected region — an instantaneous generation loss of 15,000 to 25,000 MW in ERCOT or equivalent scale in other regions. Grid frequency begins to fall immediately. Rate of change of frequency (ROCOF) is extreme — the loss of synthetic inertia from the disconnected grid-forming inverters compounds the generation loss.

T+0 to T+10 seconds: Under-frequency load shedding (UFLS) relays begin to activate as frequency falls below protection thresholds. The UFLS scheme activates load shedding across the region to arrest the frequency decline. The activation is automatic and proportional — but it was calibrated for the synchronous-generation era when large generation trips produced a different ROCOF profile. The low-inertia grid produces a ROCOF that activates UFLS at a rate and scale not anticipated by the protection settings.

T+10 to T+60 seconds: Remaining synchronous generation — gas turbines, remaining hydro — providing maximum frequency response but insufficient to arrest the decline without load shedding. UFLS activates in successive rounds across the affected interconnection area. Voltage instability developing in transmission corridors as the loss of IBR reactive support compounds the frequency event with a voltage dimension.

T+60 seconds to T+30 minutes: Controlled load shedding producing partial blackout across the affected region. Grid operators assessing the extent of the IBR disconnection — the simultaneous nature of the event across thousands of geographically dispersed facilities is initially attributed to a protection system coordination failure rather than a firmware attack. DOE, CISA, and NERC notification initiated.

T+30 minutes onward: Partial grid restoration underway. CISA and inverter OEM notification identifying the firmware activation as the cause — forensic analysis of inverter logs. Extent of compromised inverter population being mapped. Return-to-service of compromised inverters blocked until firmware can be verified or replaced. The magnitude of the compromised fleet — potentially tens of thousands of devices — makes rapid firmware replacement operationally impossible.

4.1.3 Consequence Envelope

Immediate grid consequence. A generation loss of 15,000 to 25,000 MW at near-zero inertia produces a frequency excursion whose severity depends on the UFLS scheme's ability to match load shedding to generation loss in real time. Under the low-inertia conditions that define the activation timing, the UFLS response may be insufficient to prevent a cascade into a wider blackout affecting the full interconnection. The worst case is a full interconnection blackout — the largest generation loss event in U.S. grid history.

Recovery timeline — inverter fleet. The governing recovery constraint is not the RT_HVLLC for physical component replacement. It is the firmware remediation timeline: how quickly compromised inverters can be identified, isolated, and returned to service with verified clean firmware. For a compromised population of tens of thousands of devices across hundreds of facilities, this is a months-long process — not because the physical work is complex, but because the supply chain for clean replacement inverters (from non-Chinese manufacturers) is not sized for rapid large-scale replacement. DOE's Prohibition Order authority is the legal mechanism for compelling replacement; the manufacturing capacity for rapid replacement does not exist at the required scale.

Supply chain remediation as the long-term consequence. The Scenario A event, once attributed to firmware compromise, compels a systematic assessment of the Chinese-manufactured inverter population across the full U.S. IBR fleet. The scale of that assessment — and the policy and regulatory response it triggers — constitutes a consequence that extends years beyond the immediate grid event. The manufacturing supply chain for non-Chinese inverters will need to be expanded substantially to support any systematic replacement program, a multi-year industrial policy challenge with no near-term solution.

4.2 Scenario B: Physical Cluster Attack During Grid Inertia Nadir

4.2.1 Initiating Event

Coordinated physical attack against a major utility-scale solar cluster — the Permian Basin solar corridor in ERCOT, the Desert Southwest cluster in WECC, or the Mid-Atlantic offshore wind array — during a period of maximum renewable penetration and minimum synchronous generation dispatch. The attack uses the same timing logic as the Scenario A firmware attack:

high-IBR, low-inertia grid conditions maximize the grid consequence of the physical generation loss.

The physical attack targets the collection system infrastructure — the medium-voltage cable network and collector substations that aggregate output from individual panels or turbines and deliver it to the grid interconnection point. Unlike individual panel or turbine destruction, collection system attacks can disable a large cluster with a small number of precisely selected targets: the collector substations at which all generation from a geographic area converges. A 500 MW solar cluster may have 3 to 5 collector substations; destroying two or three eliminates the majority of the cluster's output.

4.2.2 Collector Substation as the Chokepoint

The collector substation is the renewable cluster's equivalent of the pipeline pump station at a hydraulic chokepoint. It aggregates generation from a wide geographic area through a small number of physical structures. Its destruction does not require attacking individual panels or turbines — it requires attacking the point of aggregation. Collector substations for utility-scale renewable clusters are typically simple, unmanned structures with basic physical security; they are not subject to CIP-014 physical security requirements in most configurations because their individual nameplate capacity contribution falls below the BES applicability threshold.

The Scenario B consequence — loss of a major renewable cluster during a high-IBR grid condition — produces a frequency and voltage event of the same character as Scenario A but with a different recovery profile. Physical cluster destruction requires panel or turbine replacement at RT_HVLLC timelines (3 to 6 months for solar, 6 to 12 months for wind). Collector substation reconstruction is faster — 4 to 8 weeks for a standard medium-voltage collector substation — making the physical cluster attack somewhat more recoverable than the firmware attack but still producing months of generation loss during the RT_HVLLC period.

4.2.3 Offshore Wind — Maritime Compound Consequence

A physical attack against an offshore wind array combines the Scenario B collector substation attack with the maritime access constraint established in DBA-OIL-FC2. An attack on the offshore collector substation platform or the export cable landing point eliminates array output regardless of the physical condition of the individual turbines. Recovery requires offshore vessel access — subject to weather windows and vessel availability constraints — and export cable repair using specialized cable-lay vessels with lead times of 4 to 8 weeks for mobilization. The compound offshore recovery timeline — collector platform repair plus export cable repair plus turbine assessment during weather windows — produces the longest physical RT_HVLLC in the FC8 series: 12 to 24 months for a major offshore array compound attack.

4.3 Scenario C: BESS Thermal Runaway Cascade — Adversarial BMS Manipulation

4.3.1 Initiating Event

Adversarial manipulation of the battery management system at a major utility-scale BESS installation — through direct BMS firmware compromise or through SCADA-based parameter modification — designed to drive battery cells into conditions that initiate thermal runaway. The adversary has pre-positioned in the BMS control system and modifies the cell temperature and state-of-charge limits to allow the cells to charge beyond safe limits during a high-solar-generation period when grid prices incentivize maximum BESS charging. The overcharge condition — invisible to the operator because the BMS is reporting falsified cell status data — builds thermal stress in the battery modules over a period of hours.

The thermal runaway initiates in the most thermally stressed battery container when an ambient temperature spike — a hot afternoon — provides the final thermal increment. The initial thermal event propagates to adjacent containers as the Moss Landing pattern repeats: once one container exceeds the thermal runaway threshold, the heat release is sufficient to bring adjacent containers to threshold sequentially.

4.3.2 Firefighting Suppression Constraint

BESS thermal runaway fires are among the most difficult industrial fires to suppress. The lithium battery chemistry produces its own oxygen during thermal decomposition, making the fire self-sustaining regardless of external oxygen suppression. Water application risks hydrogen gas generation and electrical hazards. Specialized BESS firefighting agents — typically large quantities of water applied in a flood-cooling mode rather than a suppression mode — can prevent fire propagation to adjacent containers but cannot suppress a thermal runaway fire in an affected container. The practical consequence at the Moss Landing scale is that firefighters protect the perimeter while the affected containers burn to exhaustion.

At a facility scaled to the FC8 applicability threshold — 100 MW / 400 MWh — a full thermal runaway cascade involving the majority of battery containers produces a fire event measured in days to weeks of burn time, a site contamination consequence requiring months of remediation before rebuilding can begin, and a PFAS and toxic combustion byproduct consequence that may trigger EPA response and restrict site access.

4.3.3 Consequence Envelope: BESS Thermal Runaway

Immediate generation loss. A major BESS installation providing frequency regulation, voltage support, or peak capacity during a grid stress event is removed from service instantaneously.

Unlike a physical cluster attack where the generation loss occurs suddenly from a kinetic event, the BESS thermal runaway may develop over hours — but the fire event when it initiates produces an immediate, complete, and sustained generation loss. If the BESS installation was in active discharge supporting a grid emergency at the time of initiation, its sudden loss compounds an already-stressed grid condition.

Site remediation timeline. BESS thermal runaway produces contamination from lithium compounds, thermal decomposition byproducts, and firefighting runoff. Site remediation before rebuilding can begin: 3 to 6 months. Battery module procurement and installation: 6 to 12 months. Total RT_HVLLC from thermal runaway to return to service: 9 to 18 months, governed by the site remediation and battery module supply chain timelines.

Community and environmental consequence. BESS thermal runaway at the FC8 scale produces combustion byproducts — hydrogen fluoride from LFP battery decomposition, toxic organic compounds — that may require community evacuation and air quality monitoring. The 2023 Moss Landing event triggered community evacuation orders. At the larger scale addressed in FC8, the community consequence may extend to nearby population centers and require EPA response coordination.

5. Protective Action Thresholds

5.1 Threat Detection and Indicator Taxonomy

5.1.1 Elevated Threat Indicators

Intelligence Community reporting of adversary interest in U.S. IBR fleet firmware or inverter supply chain infrastructure. CISA notification of newly identified undisclosed communication functionality in deployed Chinese-manufactured inverters. Anomalous inverter communication activity: communication to external IP addresses not identified in the inverter's documented communication architecture; firmware update attempts through non-standard channels.

Aggregate BESS BMS data showing cell temperature or state-of-charge readings that are inconsistent across independently instrumented sensors. Pattern of collector substation physical surveillance at major solar or wind cluster aggregation points.

5.1.2 Imminent Threat Indicators

Confirmed undisclosed command received by any deployed inverter through its communication interface. Simultaneous operational anomalies at multiple IBR facilities within the same regional grid footprint inconsistent with weather or operational variation. BESS BMS data showing cell conditions inconsistent with reported state-of-charge across multiple containers at a single facility. Confirmed unauthorized access at any collector substation. Grid frequency anomaly pattern consistent with simultaneous IBR disconnection rather than a single large generator trip.

5.2 Decision Authority Framework

Facility operator level: Asset owner authority to initiate emergency disconnection of compromised or suspicious inverters, isolate BESS modules showing thermal anomalies, and notify grid operator of IBR unavailability. NERC CIP incident reporting for confirmed or suspected cyber incidents affecting inverter control systems. CISA notification for any confirmed

undisclosed inverter communication activity. Fire department notification for BESS thermal events.

Grid operator level: RTO/ISO emergency operating procedures for sudden large IBR disconnection: frequency response activation, synchronous generation emergency dispatch, and load shedding authorization. NERC event analysis notification. DOE and CISA notification for events consistent with coordinated firmware activation.

Federal coordination: CISA for inverter firmware attack attribution and response coordination. DOE CESER for supply chain emergency response and Prohibition Order authority. NERC for reliability consequence analysis and standards emergency action. FBI and DOD for confirmed state-actor firmware attack attribution. EPA for BESS thermal runaway community and environmental consequence coordination.

5.3 Federal Coordination Thresholds

Any confirmed activation of malicious firmware in deployed utility-scale inverters, regardless of scale. Simultaneous IBR disconnection affecting more than 1,000 MW in any single RTO footprint from a cause inconsistent with weather or protection system operation. BESS thermal runaway at a facility above the FC8 applicability threshold producing community evacuation or air quality emergency. Physical attack on any collector substation serving a renewable cluster above the FC8 applicability threshold. Confirmed undisclosed communication functionality in inverters representing more than 1 GW of deployed IBR capacity.

6. Recovery Architecture

6.1 The Firmware Remediation Challenge

The governing recovery constraint for Scenario A is unlike any recovery constraint in the DBA-ES series. It is not equipment procurement, regulatory authorization, or workforce availability. It is the combination of: identifying the full scope of the compromised inverter population; developing and distributing verified clean replacement firmware; and replacing or remediating tens of thousands of deployed devices across hundreds of facilities. Each of these steps faces constraints that cannot be rapidly resolved.

Identifying the compromised population requires active scanning of deployed inverters for undisclosed communication functionality — a process that requires access to every affected facility and technical capability that most asset owners do not have. DOE's National Laboratories (NREL, Sandia, PNNL) have the technical capability but not the operational scale. Developing verified clean replacement firmware requires either the original manufacturer's cooperation — which may not be forthcoming from a Chinese OEM — or the development of replacement firmware by a trusted third party. Deploying replacement firmware or physical inverter

replacement at the scale of tens of thousands of devices requires a coordinated federal response with no existing operational template.

6.2 Recovery Sequencing

Phase 1 — Immediate (0–72 hours). Grid stabilization through synchronous generation emergency dispatch and controlled load shedding. Isolation of confirmed compromised inverters from grid connection. CISA-coordinated inverter scanning protocol activation across the affected IBR fleet. DOE Prohibition Order assessment for the identified compromised device population. NERC emergency reliability standard activation. Grid operating procedures revised to account for reduced IBR availability pending firmware remediation.

Phase 2 — Short-term (72 hours to 8 weeks). Compromised inverter population mapping completed across the affected region. Trusted firmware development or alternative inverter procurement initiated. Non-Chinese inverter manufacturers contacted for emergency production capacity. Grid reliability assessment for operation with reduced IBR availability — synchronous generation emergency scheduling and demand response contracts. BESS thermal runaway site remediation initiated.

Phase 3 — Medium-term (8 weeks to recovery). Firmware remediation or inverter replacement progressively deployed across the compromised population. Sequential return to service as each facility completes remediation and firmware verification. NERC interconnection test requirements for each returning IBR. Physical cluster and BESS replacement per RT_HVLLC timelines from Table 1.

6.3 Critical Recovery Constraints

Non-Chinese inverter manufacturing capacity. The governing constraint for large-scale inverter replacement following a Scenario A event. Non-Chinese utility inverter manufacturers — primarily European (SMA, ABB/Hitachi) and American (Enphase for residential, limited utility-scale domestic production) — do not have the production capacity to replace Chinese-manufactured inverters at the scale required for the U.S. IBR fleet on any timeline measured in months. Expanding domestic and allied-nation inverter manufacturing capacity is a multi-year industrial policy initiative with no emergency shortcut. The only near-term mitigation is inverter operational restrictions — limiting dispatch of compromised-population inverters to conditions where their simultaneous disconnection would not produce a grid-destabilizing event.

BESS site remediation before rebuild. The 3 to 6 month site remediation timeline for BESS thermal runaway is independent of and precedes the battery module replacement timeline. The site cannot be rebuilt while remediation is underway; the remediation timeline is the critical path for BESS recovery. Contamination assessment, waste disposal, and regulatory clearance all gate the physical rebuild.

Offshore wind construction vessel availability. For offshore wind compound attacks, the maritime construction vessel constraint from DBA-OIL-FC2 applies: specialized offshore cable-lay and construction vessels are a globally scarce resource with long mobilization timelines. Simultaneous offshore wind and offshore petroleum infrastructure events would compete for the same vessel fleet.

Grid stability during IBR fleet remediation. Operating the grid with a large fraction of the IBR fleet isolated pending firmware remediation requires a sustained period of higher synchronous generation dispatch. In regions where synchronous generation retirement has outpaced IBR growth, this may not be achievable without emergency generation procurement or demand management. The grid reliability consequence of the Scenario A firmware attack extends beyond the immediate event through the full remediation period.

7. Gaps and Recommendations

7.1 Regulatory Gap: CIP-013 Calibration for Chinese Inverter Supply Chain Risk

NERC CIP-013 requires registered entities to assess and manage cybersecurity risks in their industrial control system supply chains. The specific risk profile of Chinese-manufactured inverters — documented undisclosed remote access functionality, state-actor control of major manufacturers, and the firmware attack vector described in Scenario A — is not explicitly addressed in CIP-013's current implementation guidance. NERC, working with CISA and DOE, should develop a CIP-013 supplement specifically addressing the Chinese inverter supply chain risk, including mandatory firmware inventory requirements, mandatory notification of any undisclosed communication functionality discovered in deployed inverters, and a defined timeline for remediation or replacement of inverters found to have undisclosed functionality.

7.2 Domestic Inverter Manufacturing Capacity

The fundamental vulnerability addressed in Scenario A cannot be resolved through cybersecurity standards alone. The root cause is the absence of a diversified inverter manufacturing base that would reduce dependence on Chinese-manufactured devices to a level below the grid-destabilizing threshold. DOE's Manufacturing USA program and the Defense Production Act provide mechanisms for supporting domestic inverter manufacturing capacity expansion. A targeted federal investment program — analogous to the CHIPS Act for semiconductors — focused on utility-scale inverter manufacturing capacity would provide the industrial base required to support large-scale inverter replacement following a Scenario A event or as a systematic risk reduction measure.

7.3 BESS Thermal Runaway Prevention and Suppression Standards

The non-adversarial and adversarial BESS thermal runaway scenarios share the same physical consequence: a fire that overwhelms available suppression resources. No mandatory federal standard currently governs BESS thermal runaway prevention, detection, or suppression at the utility scale. NFPA 855 provides installation guidance; FERC interconnection standards require protection settings. Neither addresses the adversarial BMS manipulation pathway identified in this document or the community consequence management for a large-scale thermal runaway event. A mandatory utility-scale BESS safety standard — developed by DOE in coordination with NFPA, FERC, and state fire marshals — should address BMS cybersecurity, independent cell temperature monitoring, automatic thermal runaway containment, and community emergency planning requirements.

7.4 Collector Substation Physical Security

Collector substations aggregating output from major renewable clusters are not subject to NERC CIP-014 physical security requirements in most configurations. Their low individual nameplate contribution — below BES applicability thresholds — creates a physical security gap at precisely the aggregation points whose destruction produces the highest generation loss per attack. NERC should evaluate whether collector substations serving renewable clusters above the FC8 applicability threshold should be subject to CIP-014 physical security requirements based on the aggregate generation they connect, regardless of the individual substation's own nameplate contribution.

7.5 Grid Stability Standards for Low-Inertia Operations

The Scenario A and B attacks exploit the low-inertia grid conditions that result from high IBR penetration. NERC's Reliability Standards do not currently specify minimum inertia requirements for grid operations, nor do they require grid operators to assess the frequency response consequence of simultaneous large-scale IBR disconnection under low-inertia conditions. NERC, working with the RTOs, should develop minimum inertia requirements and frequency response adequacy standards that are calibrated to the IBR penetration levels expected over the next decade — requirements that would limit the grid's exposure to the Scenario A consequence by requiring minimum synchronous generation dispatch during high-IBR-penetration periods.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial version. Establishes Design Basis Accident — Military Action scenarios for the renewable generation and battery energy storage facility class. Issued under DBA-ES Electric Sector Framework v0.7. FC designation DBA-ES-GN-FC8; covers utility-scale solar PV, onshore and offshore wind, and BESS in a single instrument.

			Applicability threshold: 200 MW aggregate for solar and wind clusters; 100 MW / 400 MWh for BESS. Three scenarios developed: Scenario A (inverter firmware attack — simultaneous grid-destabilizing behavior from compromised IBR fleet; supply chain primary with cyber activation), Scenario B (physical cluster attack during grid inertia nadir — collector substation as chokepoint; offshore wind compound maritime consequence), Scenario C (BESS thermal runaway cascade — adversarial BMS manipulation producing Moss Landing-scale fire). Dual Design Basis architecture established with four FC8-specific AI governance interaction points, including the series-first inverter firmware trustworthiness as a Bright Line precondition. RT_HVLLC values from GenMix Optimization Whitepaper v0.3 Table 1 incorporated. Chinese inverter supply chain concentration (80-85% market share) and CISA-documented undisclosed functionality established as the governing Scenario A evidentiary basis. OI-2 (DBA-EN-SF) resolved: coal, oil-fired, and renewable GenCo FCs now at v0.1. Regulatory gaps: CIP-013 calibration for Chinese inverter risk, domestic inverter manufacturing capacity, BESS thermal runaway standards, collector substation physical security, and low-inertia grid stability standards.
--	--	--	---

— END OF DOCUMENT —